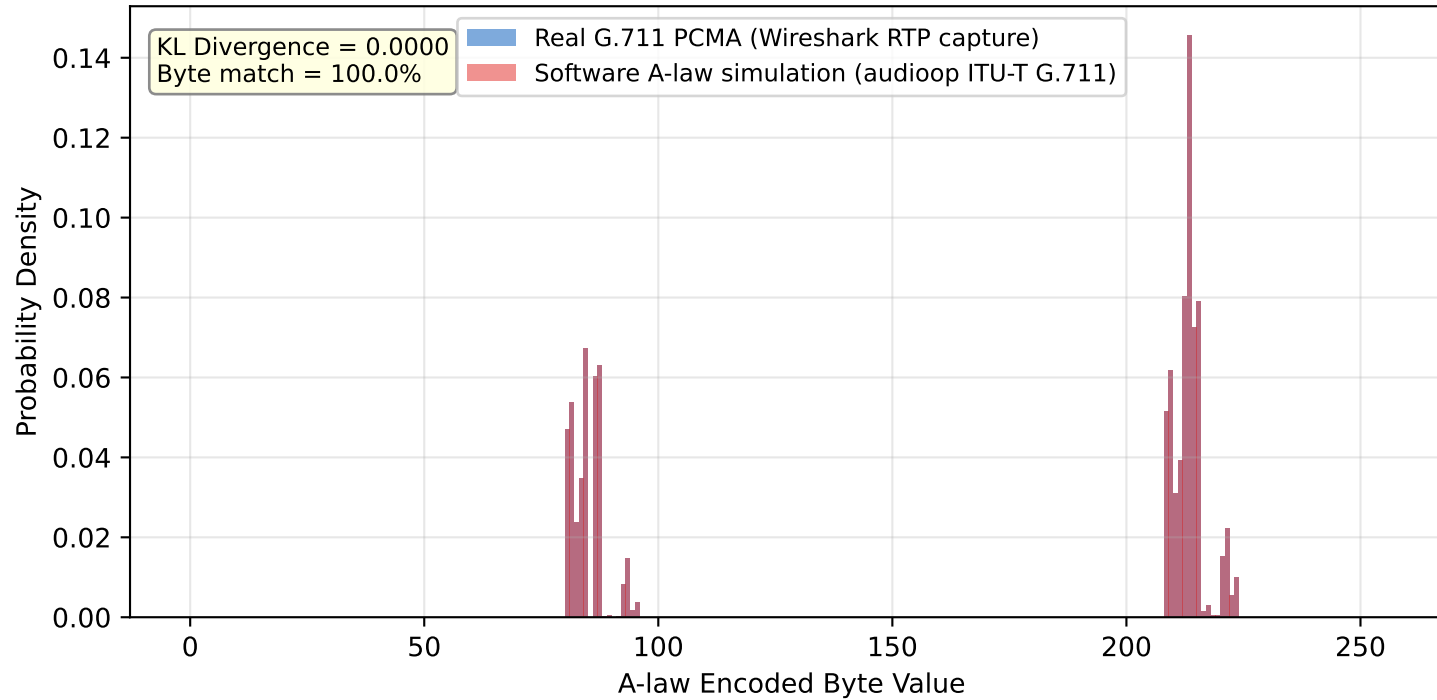


Validation: G.711 A-Law Software Simulation vs Real RTP Network Traffic (PT=8, G.711 PCMA, 8kHz, from production telephony system)

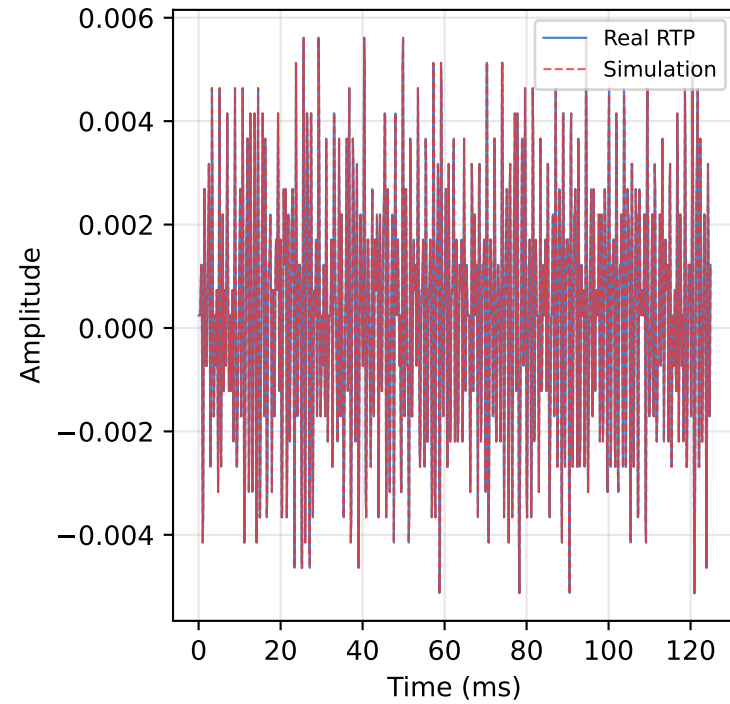
G.711 A-law Codec Output Byte Distribution



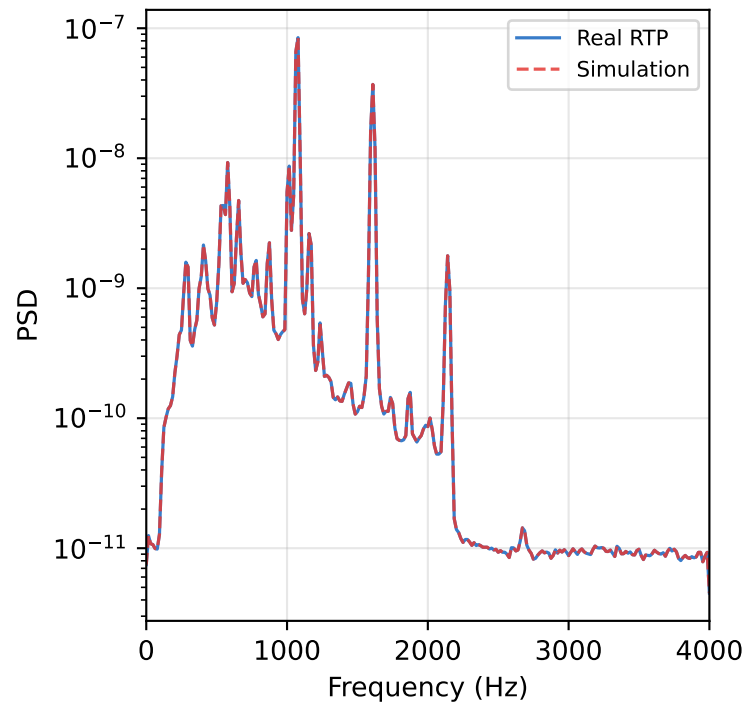
Statistical Summary

Metric	Real RTP	Simulation
Duration	14.5s	14.5s
PCM Mean	0.00030	0.00030
PCM Std	0.00232	0.00232
Byte Match	100.0%	—
KL Div	0.0000	—
KS stat	0.0000	$p \approx 1.00e+00$
Sample Rate	8000 Hz	8000 Hz
Codec	G.711 PCMA	audioop G.711

Decoded PCM Waveform (125ms window)



Power Spectral Density (0-4000 Hz bandwidth)



Q-Q Plot (alignment = statistical equivalence)

